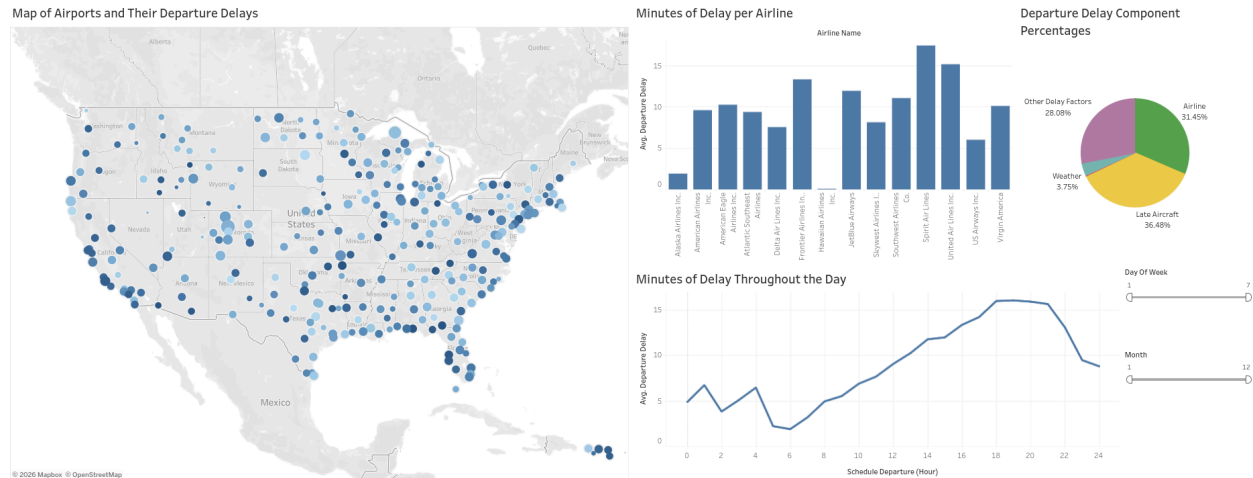


Maps and Graphs to Minimize Departure Delay

<https://public.tableau.com/app/profile/joshua.mcclintic/viz/MapsandGraphstoMinimizeDepartureDelay/Dashboard1>



Foreword:

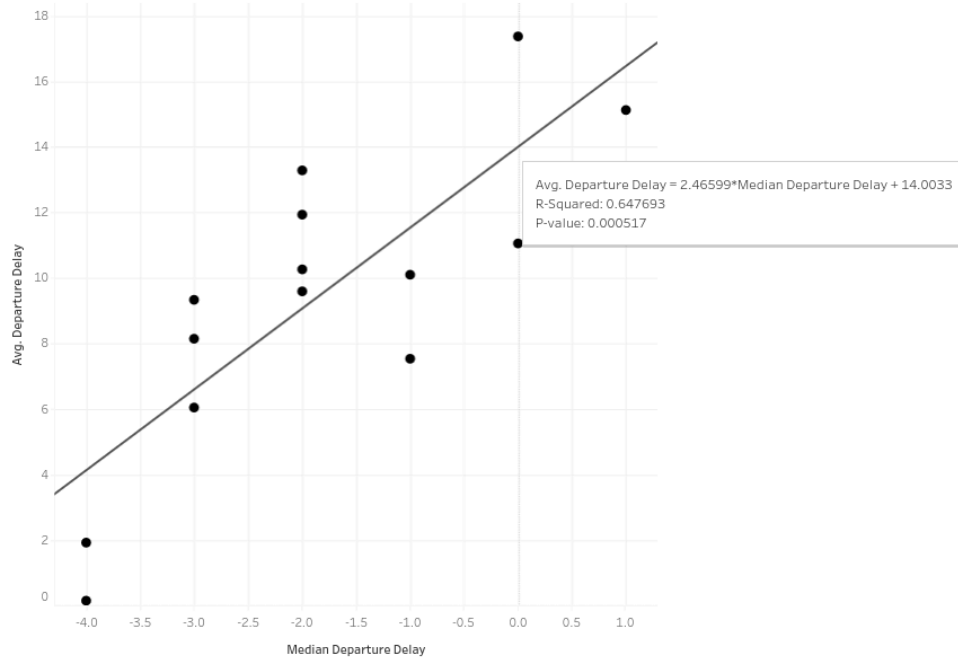
The dataset I used for the project was the “Flight-Delays” dataset. It should be noted that this dataset only contains data from 2015. An assumption is made that airports and airlines are somewhat consistent year-after-year, and the data can be used to predict with some degree of accuracy the actual delays of flights. Of course, there have almost certainly been many changes over the past decade that have significantly changed the average delays of flights. If this assumption seems particularly egregious to you, you can pretend that this was made in 2016, and therefore the data is still relevant.

Also, since this is a pdf, filters cannot be used here, but links to the Tableau pages exist as hyperlinks as well as regular links below each hyperlink.

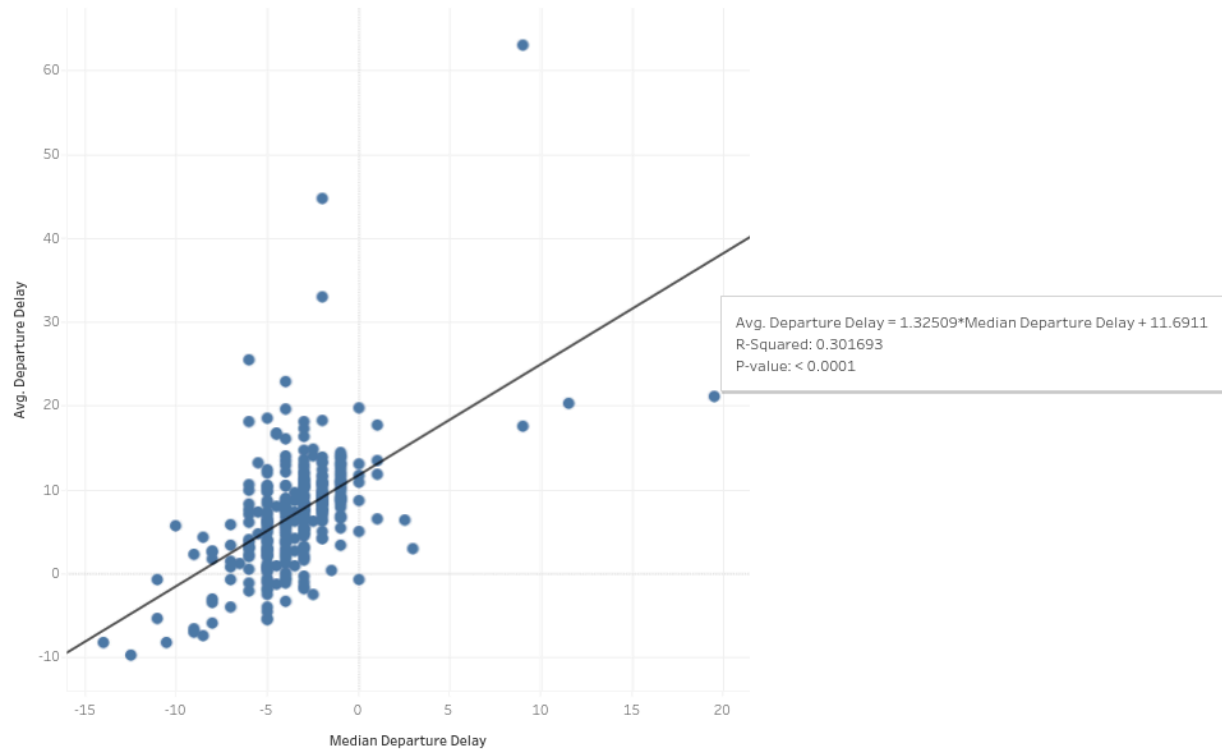
Visualizations:

There are 4 visualizations on the dashboard, all of which can be filtered by airline by clicking on a column in the “Minutes of Delay per Airline” graph, the “Month” or “Day of Week” sliders on the right, or a bubble in the map, which corresponds to a specific airport. There are also two scatter plots, named “Median vs Mean [VAR]”, which both show that there is a positive correlation between median and mean departure time for airlines and airports. The point of these graphs is to justify the use of mean, rather than median. While the median is generally a better measure of central tendency, as it is less likely to be affected by outliers, the mean here has far greater variance, and since my goal is primarily to compare different values within a variable (such as different times or airports), mean was chosen as the aggregator for delays.

Median vs Mean Airline



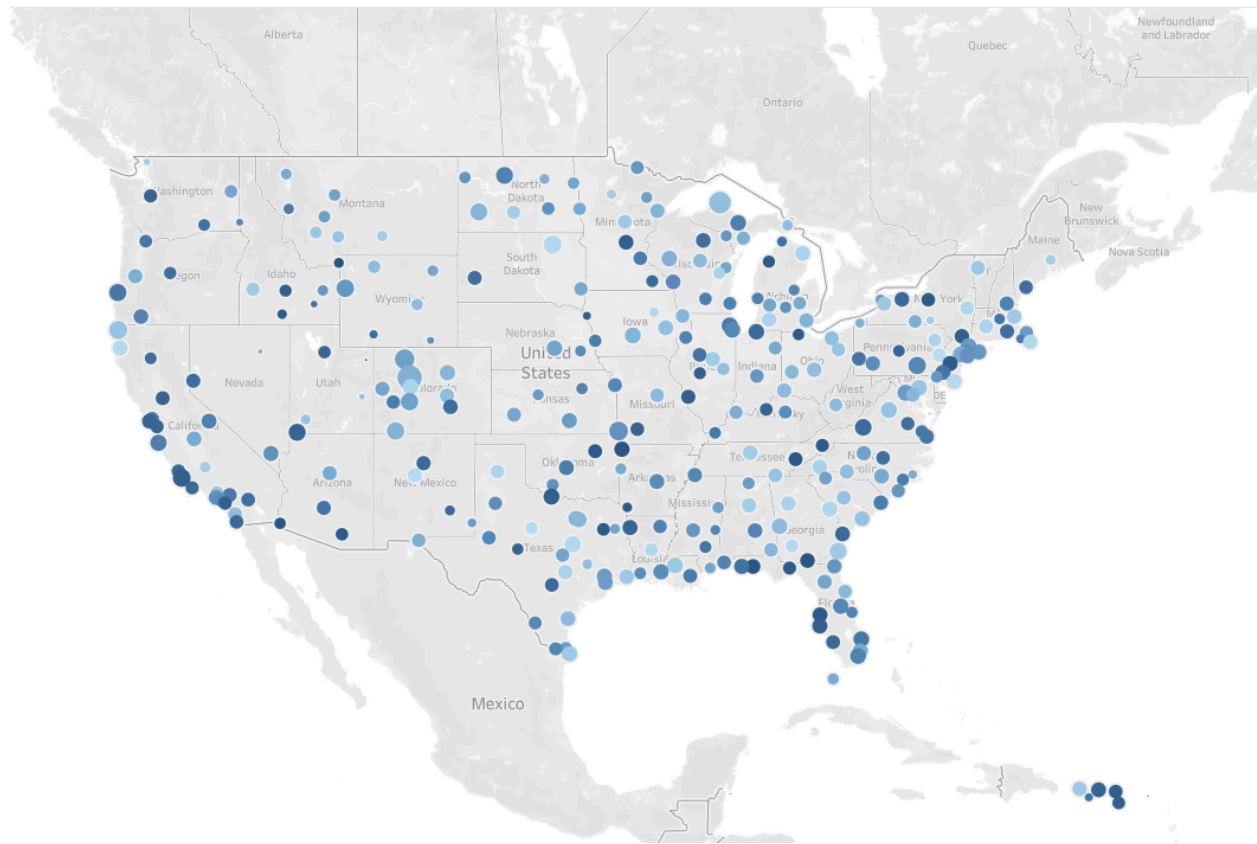
Median vs Mean Airport



Plots showing correlation between median and mean; the slope of each plot, depicted to the right of each plot, shows how much more variance the mean has compared to the median. Black is used in the first for higher contrast, as it contains sparser data.

Map of Airports and Their Departure Delays:

<https://public.tableau.com/app/profile/joshua.mcclintic/viz/MapofAirportsandTheirDepartureDelays/AirportDelaysMap>



The map is initially zoomed on the mainland US, but the dataset and the map, contain information for surrounding US territories, as well as Hawaii and Alaska, if you zoom out.

For the airport delays, a map is used to show the massive volume of data in an easy-to-read format. Because each airport only exists in one location, the location of the airport can be used to differentiate between airports. The color saturation can be used to distinguish between airports that are close together. Each bubble in the map represents an airport, and the size of the bubble represents mean departure delay. This visualization can be used to decide which airport to go to for a flight, in the case that the choice of airlines, destinations, and flight times are the same between the two.

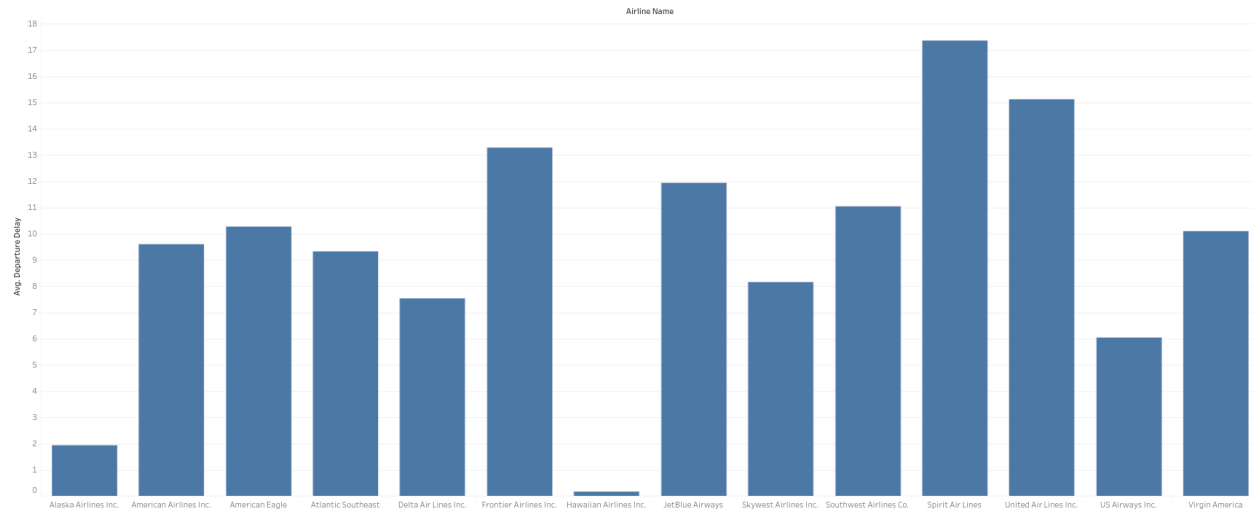
For example, if you live in central Colorado, and you plan on taking a flight on Saturday, it may be better to go to “City of Colorado Springs Municipal Airport” with an average departure delay of around -2 minutes (leaving 2 minutes early), rather than “Gunnison Crested Butte Regional Airport” with an average departure delay of over 90 minutes.

As stated earlier, you can click on a bubble in the map to change the other visualizations to only show data related to that airport.

Minutes of Delay per Airline:

<https://public.tableau.com/app/profile/joshua.mcclintic/viz/MinutesofDelayperAirline/AirlineDelays>

Minutes of Delay per Airline



For showing the difference between departure delays for airlines, a bar graph is used. This is because there are only a few different airlines, and a bar graph makes it easy to compare different airlines. Because different colors would not add information, a flat color is used throughout. Each bar is a single airline, and each y-tick is 1 minute.

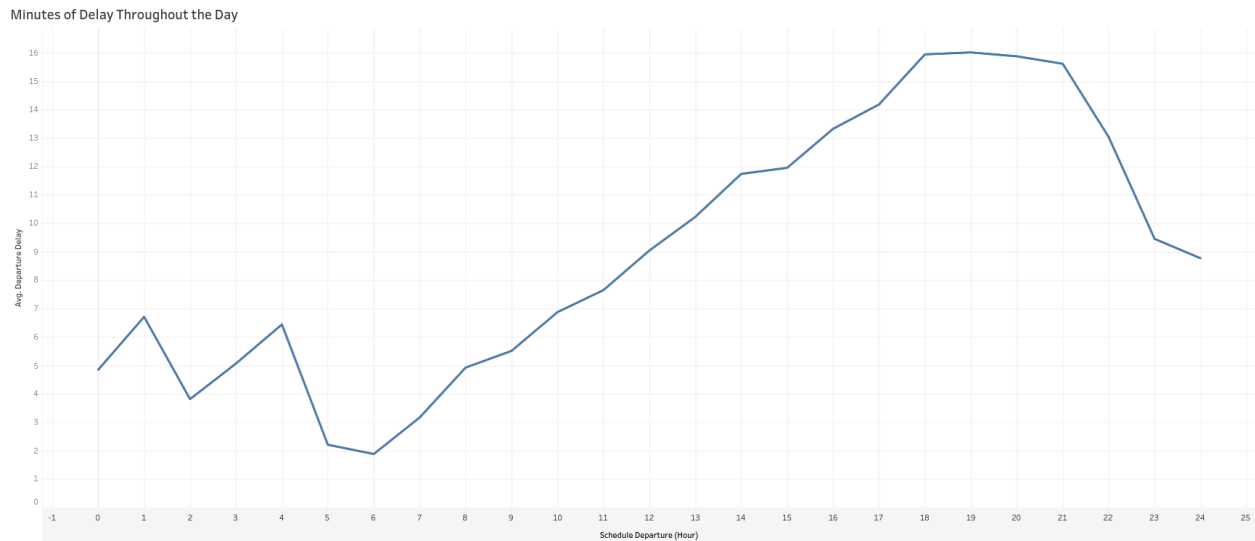
This bar graph shows the average departure delay for each airline. It is not sorted, as the order changes depending on the filter set on the graph, and keeping track of which column represents which airline as they move around is somewhat annoying. Notably, Spirit Airlines is the worst in regards to delays, and Hawaiian Airlines Inc. is the best.

As stated earlier, each bar can be selected to filter the entire dashboard to only show flights from that airline.

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Minutes of Delay Throughout the Day:

<https://public.tableau.com/app/profile/joshua.mcclintic/viz/AvgDelayThroughouttheDay/DelayThroughDay>



A line graph is used for this graph, as it works best to show how the departure delay changes over time, specifically throughout the day. Because airline and airport can be used as filters in the full dashboard, a single line is used. Also, too many lines would clutter the graph. Each X-tick represents 1 hour, and each y-tick represents 1 minute.

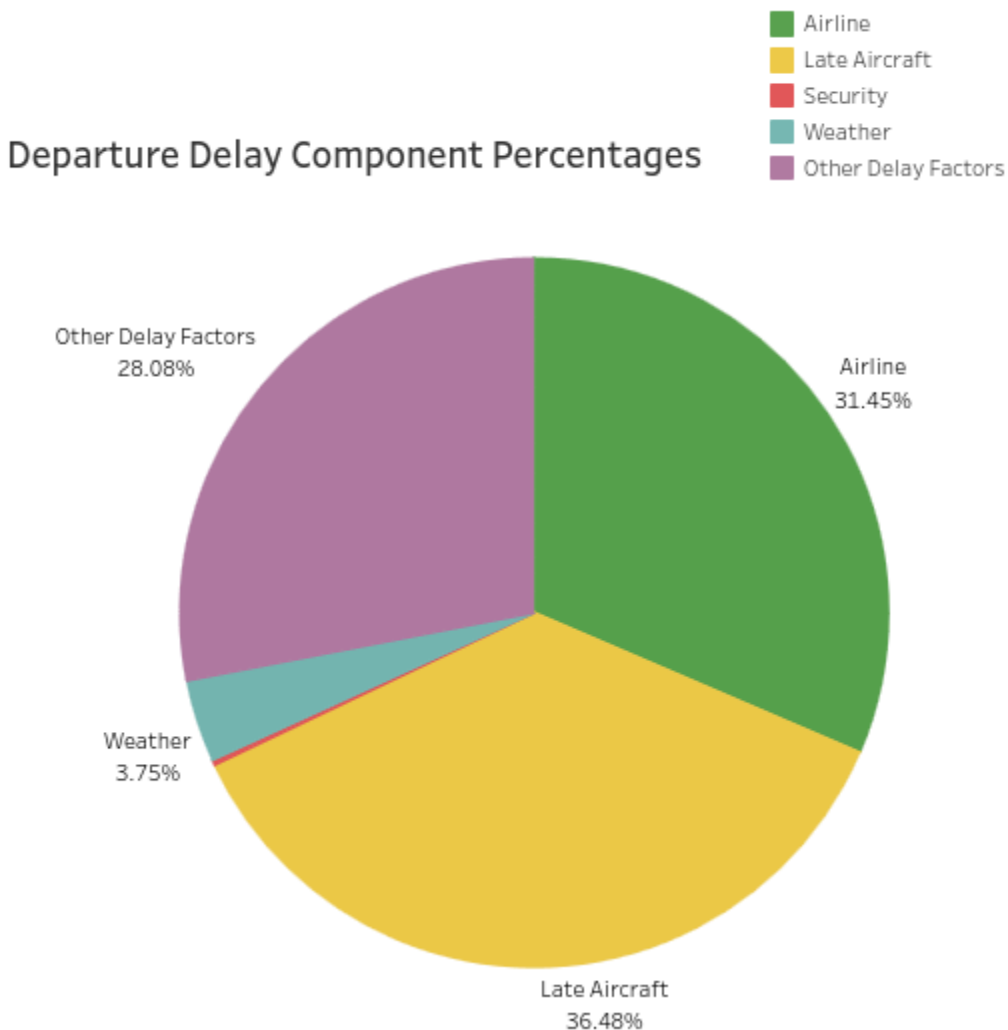
This line graph shows the average delay for any given time (represented by hours for a 24-hr clock) throughout the day. Notably, flights tend to have less delay in the morning compared to late in the day, but lowering again in the night.

While the dataset did have a column for time, the graph created with the column was hard to read. This is due to the fact that the column was a 4-digit whole number representing time. In order to make the graph readable, a new calculated field was created by dividing the 4-digit number by 100, then rounding. This gave a new 2-digit number from 0-24, which was used in the graph.

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Departure Delay Component Percentages:

<https://public.tableau.com/app/profile/joshua.mcclintic/viz/DepartureDelayComponentPercentages/DelayComponents>



A pie graph is used here to show relative sizes of the components that make up total departure delay, and pie charts are great for showing percentage sizes for small numbers of components. The colors used, based on <https://asada.website/webCVS/>, a browser app that simulates different kinds of color blindness, do not appear to be significantly impacted by color blindness.

This pie graph shows the different components of the departure delay. The dataset contains 4 different variables whose sum is usually close to the departure delay. If the departure delay is negative, these variables are typically all 0. Sometimes the departure delay is greater than the sum of the 4 components, though the dataset does not contain information on this difference. As such, a new calculated field was created, labeled on the graph as "Other Delay Factors". Unfortunately, this component takes up a large slice of the graph, so around 28% of the average delay is just "something", rather than an actual explanation.

Some notable things about this graph are that security rarely accounts for more than 1% of the total departure delay (the tiny red slice between the blue Weather and yellow Late Aircraft), weather only takes up less than 5%, and most of the delay is actually due to aircrafts being late. The second major component of delay is the airline delay, which in the [metadata on kaggle](#) is simply stated as “delay caused by the airline”. For some airports and airlines, the order of these two swaps. Importantly, this means that most of the delay is caused by either airplanes being late or by the airline, and reducing any delay would decrease late airplane delay, so the biggest actual delay factor is the airlines themselves.

References:

No references were used during this project